Student's Name: **.....**

THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION TERMINAL EXAMINATIONS - MAY, 2025

FORM THREE

SUBJECT CODE: 033 **BIOLOGY** **Time: 3:00 Hours**

INSTRUCTIONS:

- This paper consists of sections A, B and C
- Answer all questions in section A and B and any two questions in section C.
- Expect for diagrams that must be drawn in pencil all writing should be in blue or black ink

SECTION A (16 MARKS)

1. For each of the item (i – x) choose the correct answer from among the given alternatives and write its letter in the answer sheet provided
   i) Which one below represents unicellular Eukaryote?
      A. Bacteria B. Plants C. Amoeba D. Human
   ii) One of the distinctive features of kingdom fungi is possession of .....
      A. Plasma membrane b) Cytoplam C) Chitin material
      B. Cellulose material
   iii) When an animal cell is placed in the hypertonic solution, its cell membrane will be
      A. Turgid B. Haemolysed C. Plasmolysed D. Crenated
   iv) Bleeding gums is a symptoms which can be shown by a person suffering from
      A. Rickets B. Beriberi C. Pellagra D. Scuvy
   v) The element nitrogen is needed to from .....
      A. Fat B. Protein C. Starch D. Sugar
   vi) The hair like structures in moss plants which may be used to absorb water and minerals are known as .....
      A. Rhizoids B. Rhizomes C. Rhizomes D. Prothallus
   vii) The process by which ruminants brings back food into the mouth is known as ...
      A. chewing cud B. Vomiting C. Regurgitation D. Assimilation
   viii) The straghteing of the arm results ito relaxation of:-
      A. Biceps B. Skeletal muscles C. Triceps D. Smooth muscles
   ix) The group of organisms with similar feeding habits is known as .....
      A. Habitat B. Population C. Trophic level D. Community
   x) In high jump competition John fell strangely. when the stood up he could hardy maintain his body balance. which of his brain structure was damaged?
      A. Cerebrum B. Medulla oblongata C. Cochlea D. Cerebellum
2. Match the item in List a with response in list B by writing the letter of the correct response beside the item number

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LIST A	LIST B
i) Elbow joint ii) Axial skeleton iii) Muscle which never fatigue iv) Intestinal muscles v) Wrist joint vi) Mandible	A. Skull and vertebral column B. Rib cage and fore limb bones C. Gliding joints D. Fixed joints E. Biceps and skeletal muscles F. Cardiac muscles G. Ball and socket joints H. Hinge joints I. Smooth muscle J. Hold upper teeth. K. Hold lower teeth.

SECTION B: (54 MARKS)

Answer **ALL** questions in this section

3. a) What is the meaning of waste?
   b) i) What will happen if a society fail to practice proper waste disposal? 05 points
   ii) Briefly describe three basic principles of waste disposal

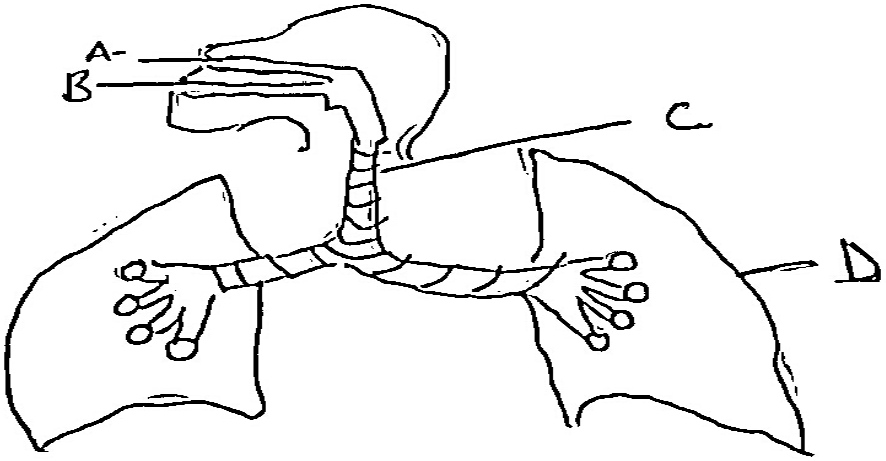
4. a) Draw a large and well-labelled diagram of an organism found in protoctista Kingdom that has both plant and animal characteristics.
   b) i) What is the name of organism drawn in 5(a) above.
   ii) Give two economic importance of kingdom protoctista.
   iii) Why the organism you mentioned above is termed as both animal and plant organism? (02 points).

5. a) Define the following terminologies as used in balance of nature.
   i) Food chain ii) Food web iii) habitat iv) community
   b) You are provided with the following organisms Grass, goat, Cow, Bacteria, man
   i. Construct two simple food chain
   ii. Construct a food web from the list of provided organisms.

6. a) Mention three factors that affect rate of diffusion
   b) i) Name two process in living things which occur by diffusion
   ii) Explain how unicellular organisms like Amoeba avoid bursting as a result of excess absorption of water by Osmosis.

7. a) Outline four characteristics of alveolus.
   b) The diagram below represents as section of human respiratory system

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i) Label the part represented A, B, C, D
ii) One can inhale through path A or B. Give reasons and state the more appropriate path
iii) Explain the effect of regular tobacco smoking on the function of the organ labeled D

8. a) While cooking in the kitchen, Madillu touched a hot pan accidentally. He moved his hand abruptly from it
   i) What is the name of process madillu did?
   ii) Define the process you mentioned above in qn(i).
   iii) Differentiate between simple reflex actions from condition reflex actions. (04 points)

   b) Draw a well labeled diagram of motor Neurone and show direction of impulse.

SECTION C: (30 MARKS)
Answer any two questions in this section

9. a) i) Define skeleton
   ii) Describe three types and three function of skeleton
   b) Explain how the human skeletal system is adapted to carry out its functions? (03 points).
10. Write an essay in a pandemic disease known as COVID – 19 using the following guidelines.
    - Causes of COVID -19
    - Ways of transmission of COVID-19 (02 points)
    - Symptoms/Signs of COVID-19 (02 points)
    - Control measures of COVID-19 (02 points)
    - Treatment of COVID-19.
11. Write an essay on mammalian blood using the following guideline
    - Meaning of blood
    - Components of mammalian blood
    - Importance of mammalian blood (04 points).